

LEARNER FILE DIVIDER

CONTENT	CHECKED
PRACTICAL MODULE	

PRACTICAL MODULES

KM ID	TITLE	Credits
PM – 01	Use hand and power tools, NQF Level 3.	22

USE HAND, POWER AND MEASURING TOOLS

PRACTICAL MODULE

Module #	671101000-PM-01
NQF Level	level 3,
Notional hours	280
Credit(s)	Cr 28
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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INTRODUCTION

Congratulations on completing the program. As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility**. Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

Acknowledgment of Receipt

I _____ (Learner) acknowledge receipt of my on the job Training Logbook on this the _____ day of _____ 20____

The process of on-the-job training has been explained to me.

Signature of Learner

Name of Assessor: _____

Signature of Assessor

Instructions to the Assessor

Congratulations on your appointment as a assessor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
- Remember the mentor is the creator of learning and exposure opportunities. You should therefore not confine the learner's exposure to this logbook alone
- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of end of each month. The Seta also requires that copies be held at the companies
- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Responsibilities of the Employer

- Creating an atmosphere conducive to learning
- Giving learners ample access to the working environment. Remember that learners should be productive employees to get practical exposure to all aspects of the transport operation as required by the learnership
- Ensure that learners, mentors and assessors attend all training required and arrange and pay travelling and accommodation costs
- Ensure availability of sufficient mentors and workplace assessors

Training Provider Responsibility

- Provide all practical learning material in electronic and/or hard copy, depending on the circumstances
 - Provide training for mentors and workplace assessors if required and provide the learning material
 - Visit employers to monitor progress and provide guidance and feedback.
 - Provide an online and telephonic support system to all mentors and learners
-

Provider Accreditation Requirements for the Module

Physical Requirements:

Demonstrate access to:

Electrical environment

categories of tools, equipment and processes in order for learners to be able to plan worksite set up for installing, wiring and connecting electrical equipment and control systems

Material include mild steel of 3mm for butt and lap welds , Butt-and lap welds on copper (maximum thickness of 8mm), Brazing is limited to mild steel only (maximum thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of 6mm).

Human Resource Requirements:

Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting, commissioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Trade Certificate.

Trainer/learner ratio 1 to 12

Workshop coach/learner ratio 1 to 4

Legal Requirements:

Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Partner (AQP)

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ)

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA)

Critical Practical Activities to be Assessed Externally for the Module

None

Exemptions

None

PRACTICAL SKILL MODULE SPECIFICATIONS

List of Practical Skill Module Specifications

671101000-PM-01	Use hand, power and measuring tools,L3,Cr28
671101000-PS-02	Plan worksite set up for installing, wiring and connecting electrical equipment and control systems,L3, Cr5
671101000-PS-03	Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems,L3,Cr3
671101000-PS-04	Install wire-ways,L4, Cr5
671101000-PS-05	Install, wire and connect electrical equipment and control systems, L4,Cr38
671101000-PS-06	Conduct pre-commission inspection (power off) and test new and existing installations, L4,Cr5
671101000-PS-07	Carry out commissioning tests,L4,Cr13
671101000-PS-08	Fault find and repair electrical control systems and electrical installations, L4, Cr22

Purpose of the Practical Skill Module 1

The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The learners will be practicing skills related to preparing and using of hand, power and measuring tools.

The learner will be required to:

PM-01-PS01: Select, use and care of engineering hand tools

PM-01-PS02: Select, use and care for power tools

PM-01-PS03: Select, use and care for electrical measuring instruments

PM-01-PS04: Perform soldering activities

PM-01-PS05: Carry out basic electric arc welding in an electrical environment

PM-01-PS06: Carry out basic gas cutting in an electrical environment

SECTION A-Guidelines for Practical Skills

PM-01-PS01: Select, use and care of engineering hand tools

Scope of Practical Skill

Given work instructions, checklists, work area, drawings, documents, templates, forms, safety and quality principles, hand saws, hammers, screw drivers, sockets, spanners, chassis punches, side cutters, pliers, wire strippers, drill bits, measuring and marking off tools, fastening tools, standard operating procedures and, statutory requirements, the learner should be able to:

- PA0101 Select engineering hand tools for serviceability.
- PA0102 Use engineering hand tools.
- PA0103 Care for engineering hand tools.

Applied Knowledge

- AK0101 Workshop procedures including housekeeping practices according to statutory requirements.
- AK0102 Techniques of maintaining hand tools
- AK0103 Techniques to correctly store hand tools

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Identify Unsafe and faulty tools and mark for repair/maintenance or replacement according to set procedures
2. Use the correct application of tools to measure and mark out steel, copper, aluminium, wood, plastics and synthetic materials.
3. Use the correct application of tools for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications.
4. Use the correct application of tools to loosen or fasten a range of different types and sizes of nuts or bolts and different types and sizes of screws as per manufacturer's specifications.
5. Use the correct application of tools to strip or cut a range of different types and sizes of electrical wire as per manufacturer's specifications.
6. Remove Sharp edges on chisel/punch type tool heads with "mushroom effect" to ensure that they can be used safely; then tips with burs or chips are sharpened for effective use.
7. Fill Flat Screw driver tips with burs or chips and prepare to a manufacturer's specification ensuring that they are serviceable
8. Lubricate the adjusting/moving mechanisms of shifting spanners, gas pump pliers, vice grip, dividers, crimping tools, stilson wrench to ensure free movement and serviceability.
9. Clean Engineering files cutting edge using a wire brush to ensure serviceability. Scribes, dividers and centre punches were sharpened to ensure serviceability.

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Identify Unsafe and faulty tools and mark for repair/maintenance or replacement according to set procedures			
2. Use the correct application of tools to measure and mark out steel, copper, aluminium, wood, plastics and synthetic materials.			
3. Use the correct application of tools for cutting steel, copper,			

aluminium, wood, plastics and synthetic materials as per manufacturers specifications.			
4. Use the correct application of tools to loosen or fasten a range of different types and sizes of nuts or bolts and different types and sizes of screws as per manufacturer's specifications.			
5. Use the correct application of tools to strip or cut a range of different types and sizes of electrical wire as per manufacturer's specifications.			
6. Remove Sharp edges on chisel/punch type tool heads with "mushroom effect" to ensure that they can be used safely; then tips with burs or chips are sharpened for effective use.			
7. Fill Flat Screw driver tips with burs or chips and prepare to a manufacturer's specification ensuring that they are serviceable			

8. Lubricate the adjusting/moving mechanisms of shifting spanners, gas pump pliers, vice grip, dividers, crimping tools, stilson wrench to ensure free movement and serviceability.			
9. Clean Engineering files cutting edge using a wire brush to ensure serviceability. Scribes, dividers and centre punches were sharpened to ensure serviceability.			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

SECTION B-Guidelines for Practical Skills

PM-01-PS02: Select, use and care for power tools

Scope of Practical Skill

Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, the learner should be able to:

PA0201 Select a power tool pertaining to specific job requirements.

PA0202 Use fixed power tools.

PA0203 Use portable power tools.

PA0204 Care for and store power tools and their accessories.

Applied Knowledge

AK0201 Workshop procedures including housekeeping practices according to statutory requirements.

AK0202 Safety practices relating to the use of power tools including the use of personal protective equipment, electrical and fire protection

AK0203 Types and uses of fixed and portable power tools

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Select Power tools according to the task requirements.
2. Carry out Pre-operational check in terms of safety requirements and function.
3. Recognise Hazards associated with the use of power tools and take necessary precautions according to workshop procedures.
4. Identify and report Faulty and unsafe tools in accordance with standard operating procedures
5. Select and use Personal safety equipment according to tool requirement.
6. Select and set up Fixed and portable power tools according to job requirements.
7. Utilise all guards and securing mechanisms in terms of job requirements.
8. Use Fixed and portable power tools safely and in accordance with manufacturer's specifications.
9. Inspect, clean and lubricate Power tools after use according to workshop practices and/or manufactures specifications.
10. Secure and repair loose items and minor defects in accordance to workshop procedures.
11. Store Portable power tools and accessories in accordance to workshop procedures.

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Select Power tools according to the task requirements.			
2. Carry out Pre-operational check in terms of safety requirements and function.			
3. Recognise Hazards associated with the use of power tools and take necessary precautions according to workshop procedures.			
4. Identify and report Faulty and unsafe tools in accordance with			

standard operating procedures			
5. Select and use Personal safety equipment according to tool requirement.			
6. Select and set up Fixed and portable power tools according to job requirements.			
7. Utilise all guards and securing mechanisms in terms of job requirements.			
8. Use Fixed and portable power tools safely and in accordance with manufacturer's specifications.			
9. Inspect, clean and lubricate Power tools after use according to workshop practices and/or manufactures specifications.			
10. Secure and repair loose items and minor defects in accordance to workshop procedures.			
11. Store Portable power tools and accessories in accordance to workshop procedures.			

GENERAL COMMENTS OF OBSERVER:

Date.....

Time started.....

Time
completed.....**ASSESSOR NAME****ASSESSOR SIGNATURE****LEARNER
(SIGNATURE)**

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SECTION C-Guidelines for Practical Skills

PM-01-PS03: Select, use and care for electrical measuring instruments

Scope of Practical Skill

Given work instructions, work area, documents, multimeter (ohms, volts, amp`s), insulation tester, clip on ammeter, earth leakage polarity tester, oscilloscope standard operating procedures and, statutory requirements the learner should be able to:

- PA0301 Identify and select portable electrical measuring instruments.
- PA0302 Use and interpret portable electrical measuring instrument readings.
- PA0303 Care for portable electrical measuring instruments.

Applied Knowledge

- AK0301 Handling procedures related to portable measuring instrument.
- AK0302 Basic operating principles of portable electrical measuring instruments.
- AK0303 Types, applications and functions of portable electrical measuring instruments.

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Interpret Job Instructions correctly and sequence of operation is determined.
2. Select Portable electrical measuring instruments correctly according to workshop procedures and to meet safety and job requirements
3. Identify and mark Unsafe and faulty measuring instruments for repair or replacement.
4. Read Portable measuring instruments correctly and record voltage, amperage, and resistance readings as per workshop procedures.
5. Check Portable electrical equipment for correct operation and functionality.
6. Set up measuring instruments for application.
7. Record Results on appropriate documentation.
8. Interpret Results accurately against the specifications of the job requirements.
9. Use Electrical measuring instruments in accordance with their specifications.
10. Store electrical measuring instruments in accordance with manufactures specifications and workshop standards.

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Interpret Job instructions correctly and sequence of operation is determined.			
2. Select Portable electrical measuring instruments correctly according to workshop procedures and to meet safety and job requirements			
3. Identify and mark Unsafe and faulty measuring instruments for repair or replacement.			
4. Read Portable measuring instruments correctly and			

record voltage, amperage, and resistance readings as per workshop procedures.			
5. Check Portable electrical equipment for correct operation and functionality.			
6. Set up measuring instruments for application.			
7. Record Results on appropriate documentation.			
8. Interpret Results accurately against the specifications of the job requirements.			
9. Use Electrical measuring instruments in accordance with their specifications.			
10. Store electrical measuring instruments in accordance with manufactures specifications and workshop standards.			
GENERAL COMMENTS OF OBSERVER:			

Date.....	Time started.....	Time completed.....
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)

SECTION D-Guidelines for Practical Skills

PM-01-PS04: Perform soldering activities

Scope of Practical Skill

Given work instructions, work area, checklists, soldering wire, soldering iron/stations, solder sucker, accessories, standard operating procedures and statutory requirements the learner should be able to:

- PA0401 Solder various wire joints
- PA0402 Solder components on a PC or Vero board
- PA0403 De-solder components

Applied Knowledge

- AK0401 Soldering techniques and applications
 - AK0402 Safety practices relating to the use of soldering equipment including the use of personal protective equipment.
 - AK0403 De-soldering methods
-

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Inspect Work area for safe working conditions and corrective action is taken where required.
2. Select Applicable soldering equipment as required by task.
3. Correct select Soldering material as required by the task.
4. Clean all connections from any dirt or oxidation.
5. Tinning of wire and connections was done according to manufacturer's specifications.
6. Solder Connections according to set specifications and techniques.
7. Lay out Components on the circuit board according to the circuit diagram.
8. Solder Components according to soldering techniques.

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Inspect Work area for safe working conditions and corrective action is taken where required.			
2. Select Applicable soldering equipment as required by task.			
3. Correct select Soldering material as required by the task.			
4. Clean all connections from any dirt or oxidation.			
5. Tinning of wire and connections was done according to manufacturer's specifications.			

6. Solder Connections according to set specifications and techniques.			
7. Lay out Components on the circuit board according to the circuit diagram.			
8. Solder Components according to soldering techniques.			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

SECTION E-Guidelines for Practical Skills

PM-01-PS05: Carry out basic electric arc welding in an electrical environment

Given work instructions, work area, documents, portable electric arc welding machines, welding cables., electrode holder, earth clamp, chipping hammer, electrodes (welding rods), metal surface cleaning materials (files, wire brush, emery cloth, rags), Arc welding helmets, welding screens, leather gloves , aprons and standard operating procedures and statutory requirements the learner should be able to:

- PA0501 Prepare the electric arc welding equipment for the task.
- PA0502 Prepare the metal surfaces for electric arc welding.
- PA0503 Apply basic electric arc welding techniques.

Applied Knowledge

- AK0501 Names, applications and functions of electric arc welding machines and equipment.
- AK0502 Handling procedures related to electric arc welding equipment.
- AK0503 Application of various types of electrodes and current settings during the electric arc welding process.
- AK0504 Safe work procedures and workshop practices.
- AK0505 Workshop procedures including housekeeping practices according to statutory requirements.

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Obtain and inspect the applicable safety equipment as per safe work procedures and statutory requirements.
2. Set up the electric arc welding equipment and check cable connections for tightness to ensure effective operation as per safe work procedures.
3. Correctly apply the cleaning materials for preparing the metal surfaces to be welded as per task requirements.
4. Wear the appropriate personal protective equipment as per safe work procedures.
5. Bevel the edges of the metals welded correctly where needed as per specific task requirements.
6. Clean the contact surfaces of the earth clamp, metals earthed and the electrode holder to ensure proper current flow during the welding process.
7. Correctly select the required current setting on the welding machine for electric arc welding according to application requirements.
8. Carry out the joining of metals by means of electric arc welding was and position correct electrode positioning and movement according to application requirements.
9. Cool down Hot metal surfaces and remove all slag from the welding joints as per safe work procedures.
10. Correctly store Electric arc welding machine, equipment and materials as per applicable safety standards.
11. Clean the work area and dispose waste materials according to local housekeeping standards.
12. Clean and store Safety equipment according to statutory requirements.

OBSERVATION EVALUATION

NAME:	EVALUATION CHECKLIST
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SURNAME: COMPANY: ID:	DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Obtain and inspect the applicable safety equipment as per safe work procedures and statutory requirements.			
2. Set up the electric arc welding equipment and check cable connections for tightness to ensure effective operation as per safe work procedures.			
3. Correctly apply the cleaning materials for preparing the metal surfaces to be welded as per task requirements.			
4. Wear the appropriate personal protective equipment as per safe work procedures.			

5. Bevel the edges of the metals welded correctly where needed as per specific task requirements.			
6. Clean the contact surfaces of the earth clamp, metals earthed and the electrode holder to ensure proper current flow during the welding process.			
7. Correctly select the required current setting on the welding machine for electric arc welding according to application requirements.			
8. Carry out the joining of metals by means of electric arc welding was and position correct electrode positioning and movement according to application requirements.			
9. Cool down Hot metal surfaces and remove all slag from the welding joints as per safe work procedures.			
10. Correctly store Electric arc welding machine,			

equipment and materials as per applicable safety standards.			
11. Clean the work area and dispose waste materials according to local housekeeping standards.			
12. Clean and store Safety equipment according to statutory requirements.			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

SECTION F-Guidelines for Practical Skills

PM-01-PS06: Carry out basic gas cutting in an electrical environment

Scope of Practical Skill

Given work instructions, work area, documents, Oxygen-and acetylene gas cylinders and keys, gas regulators and flashback arrestors, rubber gas hoses and clamps, cutting torches and nozzles, nozzle cleaning equipment, spark lighter, , metal surface cleaning materials (wire brush, emery cloth, chemicals), gas welding goggles, leather aprons and boot protectors, safety gloves and overalls the learner should be able to:

PA0601 Prepare for gas cutting.

PA0602 Apply basic gas cutting process.

Applied Knowledge

AK0601 Names, applications and functions of gas cutting equipment.

AK0602 Handling procedures related to gas cutting equipment.

AK0604 Safe work procedures and workshop practices.

AK0605 Workshop procedures including housekeeping practices according to statutory requirements.

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Obtain and inspect the applicable safety equipment as per safe work procedures and statutory requirements.
2. Set up the gas cutting equipment and check for gas leaks and correct cylinder pressures according to safe work procedures and industry requirements.
3. Prepare the relevant cutting material limited to mild steel only (maximum thickness of 6mm), required for the task to be ready and available, and select nozzles correctly to match the application.
4. Handle the gas cylinders were handled safely as per safe work procedures and industry requirements.
5. Maintain Safety and proper workplace practices at all times.
6. Correctly set the oxygen and acetylene gas pressures on the regulator for gas cutting according to application requirements.
7. Lit up the gas torch in the correct sequence and the gas mixture set as per safe work procedures and industry requirements.
8. Cut the metal with the gas cutting torch and the correct nozzle position- and movement was demonstrated according to application requirements.
9. Cool down Hot metal surfaces as per safe work procedures.
10. Close Gas cylinder valves and gas pressure in the hoses released as per safe work procedures.
11. Correctly store Gas cutting equipment and materials as per applicable safety standards.
12. Clean the work area and dispose waste materials according to local housekeeping standards.

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY:	EVALUATION CHECKLIST DATE:
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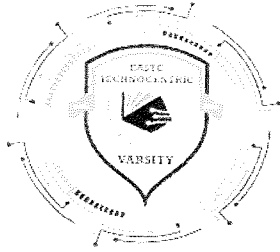
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ABILITY OF THE LEARNER TO:			
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9. Cool down Hot metal surfaces as per safe work procedures.			
10. Close Gas cylinder valves and gas pressure in the hoses released as per safe work procedures.			
11. Correctly store Gas cutting			

equipment and materials as per applicable safety standards.			
12. Clean the work area and dispose waste materials according to local housekeeping standards.			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

LOGBOOK				
Date	Assignment task	Start	Finish	Total Hours
Date	Learner signature	Date	Assessor signature	



LEARNER FILE DIVIDER

CONTENT	CHECKED
PRACTICAL MODULE	

PRACTICAL MODULES

KM ID	TITLE	Credits
PM – 02	Plan and prepare the process for the wiring, connection, testing, inspecting, commissioning and maintaining electrical installations and control systems, NQF Level 3.	5

**PLAN AND PREPARE THE PROCESS FOR THE
WIRING AND CONNECTION, TESTING,
INSPECTING, COMMISSIONING AND
MAINTAINING ELECTRICAL INSTALLATION AND
CONTROL SYSTEMS**

PRACTICAL MODULE

Module #	671101000-PS-02
NQF Level	level 3,
Notional hours	50
Credit(s)	Cr 5
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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INTRODUCTION

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You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility**. Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

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Signature of Learner

Name of Assessor: _____

Signature of Assessor

Instructions to the Assessor

Congratulations on your appointment as a assessor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
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- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of end of each month. The Seta also requires that copies be held at the companies
- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Responsibilities of the Employer

- Creating an atmosphere conducive to learning
- Giving learners ample access to the working environment. Remember that learners should be productive employees to get practical exposure to all aspects of the transport operation as required by the learnership
- Ensure that learners, mentors and assessors attend all training required and arrange and pay travelling and accommodation costs
- Ensure availability of sufficient mentors and workplace assessors

Training Provider Responsibility

- Provide all practical learning material in electronic and/or hard copy, depending on the circumstances
 - Provide training for mentors and workplace assessors if required and provide the learning material
 - Visit employers to monitor progress and provide guidance and feedback.
 - Provide an online and telephonic support system to all mentors and learners
-

Provider Accreditation Requirements for the Module

Physical Requirements:

Demonstrate access to:

Electrical environment

categories of tools, equipment and processes in order for learners to be able to plan worksite set up for installing, wiring and connecting electrical equipment and control systems

Human Resource Requirements:

Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting, commissioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Trade Certificate.

Trainer/learner ratio 1 to 12

Workshop coach/learner ratio 1 to 4

Legal Requirements:

Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality

Partner (AQP)

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA)

Requirements

Critical Practical Activities to be Assessed Externally for the Module

None

Exemptions

None

PRACTICAL SKILL MODULE SPECIFICATIONS

List of Practical Skill Module Specifications

671101000-PM-01	Use hand, power and measuring tools, L3, Cr28
671101000-PS-02	Plan worksite set up for installing, wiring and connecting Electrical equipment and control systems,L3, Cr5
671101000-PS-03	Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems,L3,Cr3
671101000-PS-04	Install wire-ways,L4, Cr5
671101000-PS-05	Install, wire and connect electrical equipment and control systems, L4,Cr38
671101000-PS-06	Conduct pre-commission inspection (power off) and test new and existing installations, L4,Cr5
671101000-PS-07	Carry out commissioning tests,L4,Cr13
671101000-PS-08	Fault find and repair electrical control systems and electrical installations, L4, Cr22

Purpose of the Practical Skill Module 2

Purpose of the Practical Skill Module

The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The learners will be practicing skills related to planning worksite set up for installation, wiring and connection of electrical equipment and control systems

The learner will be required to:

PM-02-PS01: Undertake risk assessment in accordance with all statutory requirements

PM-02-PS02: Read and interpret electrical drawings

PM-02-PS03: Plan the worksite set-up

SECTION A-Guidelines for Practical Skills

PM-02-PS01: Undertake a preliminary risk assessment

Scope of Practical Skill

Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, the learner should be able to:

- PA0101 Conduct a preliminary risk assessment.
- PA0102 Identify the hazards.
- PA0103 Relate the risk to the hazard.
- PA0104 Conduct mitigation of the risk.

Applied Knowledge

- AK0101 Safety procedures
- AK0102 Statutory requirements for hazardous conditions/material/substance.
- AK0103 Risk assessment procedure for specific context
- AK0104 Inspection procedure
- AK0105 Recording/reporting procedure
- AK0106 Approval procedures

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Carry out Risk assessments, including work related hazards and ergonomics, and complete reports as per statutory requirements.
 2. Observe Evidence of mitigation in accordance with standard operating procedures
 3. Complete approved work permit or other documentation to reflect risk assessment conducted in accordance with approval procedures
-

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Carry out Risk assessments, including work related hazards and ergonomics, and complete reports as per statutory requirements.			
2. Observe Evidence of mitigation in accordance with standard operating procedures			
3. Complete approved			

work permit or other documentation to reflect risk assessment conducted in accordance with approval procedures			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

SECTION B-Guidelines for Practical Skills

PM-02-PS02: Read and interpret electrical drawings

Scope of Practical Skill

Given work instructions, work area, case study, documents, templates, the learner should be able to:

- PA0201 Identify signs and symbols
- PA0202 Read and interpret electrical drawings

Applied Knowledge

- AK0201 Methods of interpreting scales
- AK0202 Techniques of interpreting diagrams
- AK0203 Procedures for evaluating the work site.

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Correctly identify signs and symbols were according to the IEC(International Electrical compliance standards)
 2. Correctly interpret electrical diagrams, using schematics, plans, wiring and layout representations according to worksite requirements.
-

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY ID	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Correctly identify signs and symbols were according to the IEC(International Electrical compliance standards)			
2. Correctly interpret electrical diagrams, using schematics, plans, wiring and layout representations according to worksite requirements.			
GENERAL COMMENTS OF OBSERVER:			

Date.....	Time started	Time completed.....
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)

SECTION C-Guidelines for Practical Skills

PM-02-PS03: Plan the worksite set-up

Scope of Practical Skill

Given work instructions, work area, case study, documents, the learner should be able

to: PA0201 Compile a list of tools, material and equipment required

PA0202 Compile a list of human resources required

PA0203 Compile a work flow diagram

Applied Knowledge

AK0201 Knowledge of tools, material and equipment required to perform
various functions within the worksite

AK0202 Knowledge of human resources required to perform various functions within the
worksite

AK0203 Analytic techniques for interpreting work instructions

AK0204 Knowledge of workflow processes

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

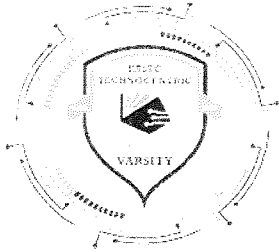
1. List the relevant tools, material and equipment and record quantities in accordance with the task specification
 2. List the correct human resources that meet the job specification
 3. Analyse the work flow / work order correctly and apply in accordance with the task specification
-

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. List the relevant tools, material and equipment and record quantities in accordance with the task specification			
2. List the correct human resources that meet the job specification			
3. Analyse the work flow / work order correctly and apply in accordance with the task specification			
GENERAL COMMENTS OF OBSERVER:			

Date.....	Time started.....	Time completed.....
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)



LEARNER FILE DIVIDER

CONTENT	CHECKED
PRACTICAL MODULES	

PRACTICALS MODULES

KM ID	TITLE	Credits
PM - 3	Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, NQF Level 3.	3

PREPARE WORKSITE SET UP FOR INSTALLING, WIRING AND CONNECTING ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS

PRACTICAL MODULE

Module #	671101000-PM-03
NQF Level	level 3,
Notional hours	300
Credit(s)	Cr 30
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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INTRODUCTION

Congratulations on completing the program. As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility**. Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

Acknowledgment of Receipt

I _____ (Learner) acknowledge receipt of my on the job Training Logbook
on this the _____ day of _____ 20____

The process of on-the-job training has been explained to me.

Signature of Learner

Name of Assessor: _____

Signature of Assessor

Instructions to the Assessor

Congratulations on your appointment as a assessor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
- Remember the mentor is the creator of learning and exposure opportunities. You should therefore not confine the learner's exposure to this logbook alone
- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of end of each month. The Seta also requires that copies be held at the companies
- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Responsibilities of the Employer

- Creating an atmosphere conducive to learning
- Giving learners ample access to the working environment. Remember that learners should be productive employees to get practical exposure to all aspects of the transport operation as required by the learnership
- Ensure that learners, mentors and assessors attend all training required and arrange and pay travelling and accommodation costs
- Ensure availability of sufficient mentors and workplace assessors

Training Provider Responsibility

- Provide all practical learning material in electronic and/or hard copy, depending on the circumstances
 - Provide training for mentors and workplace assessors if required and provide the learning material
 - Visit employers to monitor progress and provide guidance and feedback.
 - Provide an online and telephonic support system to all mentors and learners
-

Provider Accreditation Requirements for the Module

Physical Requirements:

Demonstrate access to:

Electrical environment

categories of tools, equipment and processes in order for learners to be able to plan worksite set up for installing, wiring and connecting electrical equipment and control systems

Human Resource Requirements:

Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting, commissioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Trade Certificate.

Trainer/learner ratio 1 to 12

Workshop coach/learner ratio 1 to 4

Legal Requirements:

Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Partner (AQP)

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) requirements

Critical Practical Activities to be Assessed Externally for the Module

None

Exemptions

None

PRACTICAL SKILL MODULE SPECIFICATIONS

List of Practical Skill Module Specifications

671101000-PM-01	Use hand, power and measuring tools,L3,Cr28
671101000-PS-02	Plan worksite set up for installing, wiring and connecting electrical equipment and control systems,L3, Cr5
671101000-PS-03	Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems,L3,Cr3
671101000-PS-04	Install wire-ways,L4, Cr5
671101000-PS-05	Install, wire and connect electrical equipment and control systems, L4,Cr38
671101000-PS-06	Conduct pre-commission inspection (power off) and test new and existing installations, L4,Cr5
671101000-PS-07	Carry out commissioning tests,L4,Cr13
671101000-PS-08	Fault find and repair electrical control systems and electrical installations, L4, Cr22

Purpose of the Practical Skill Module 3

The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The learners will be practicing skills related to preparing worksite set up for installation, wiring and connection of electrical equipment and control systems

The learner will be required to:

PM-03-PS01: Procure resources

PM-03-PS02: Prepare and transport resources

PM-03-PS03: Prepare work site, equipment, tools, consumables and materials

SECTION A-Guidelines for Practical Skills

PM-03-PS01: Procure resources

Scope of Practical Skill

Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, the learner should be able to:

PA0101 Obtain human resources required.

PA0102 Requisition (obtain) tools and equipment required.

PA0103 Requisition (obtain) material and consumables required.

PA0104 Determine functionality of tools and testing equipment

Applied Knowledge

AK0101 Procurement procedures

AK0102 Resource determination methods

AK0103 Techniques and procedures for requisitioning

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Determine and list the Human resources to meet the specific needs of the activity and are fit for purpose
2. Determine and procure Tools and testing instruments required to meet the specific needs of the activity and are fit for purpose
3. Procure Equipment to meet the specific needs of the activity
4. Procure Material requirements to meet the specific needs of the activity and are fit for purpose
5. Procure Consumable requirements to meet the specific needs of the activity and are fit for purpose

Attach evidence of receipts, invoices etc.

PRODUCT EVALUATION

NAME: SURNAME: COMPANY: ID: Learner Attached evidence of procurements receipts, invoices etc.	EVALUATION CHECKLIST DATE: TIME:
--	---

EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Determine and list the Human resources to meet the specific needs of the activity and are fit for purpose			
2. Determine and procure Tools and testing instruments required to meet the specific needs of the activity and are fit for purpose			
3. Procure Equipment to			

meet the specific needs of the activity			
4. Procure Material requirements to meet the specific needs of the activity and are fit for purpose			
5. Procure Consumable requirements to meet the specific needs of the activity and are fit for purpose			
Learner Attached evidence of procurements receipts, invoices etc.			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

SECTION B-Guidelines for Practical Skills

PM-03-PS02: Prepare and move resources

Scope of Practical Skill

Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, the learner should be able to:

PA0201 Load and off load resources

PA0202 Procure type and size of moving equipment

PA0203 Stack and secure tools, material and equipment

PA0203 Store tools, material and equipment on site

Applied Knowledge

AK0201 Basic lifting techniques

AK0202 Use of material safety data sheet

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Load and off load resources
2. Stack and secure tools, material and equipment
3. Store tools, material and equipment on site

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Load and off load resources			
2. Stack and secure tools, material and equipment			
3. Store tools, material and equipment on site			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

SECTION C-Guidelines for Practical Skills

PM-01-PS03: Select, use and care for electrical measuring instruments

Scope of Practical Skill

Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, the learner should be able to:

- PA0301 Undertake safety checks
- PA0302 Source temporary electrical supplies for use
- PA0303 Position equipment for installation

Applied Knowledge

- AK0301 Lock out procedures
- AK0302 Relevant subsections of SANS 10142-1
- AK0303 Installation requirements
- AK0304 Site specific rules and regulations

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Complete a safety report that contain all relevant areas of the installation
 2. Correctly position Equipment and components according to the installation requirements
 3. Correctly apply the relevant subsections of SANS 10142-1
-

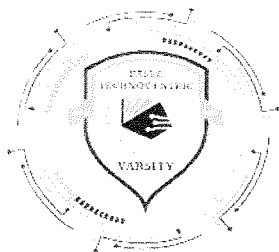
OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Complete a safety report that contain all relevant areas of the installation			
2. Correctly position Equipment and components according to the installation requirements			
3. Correctly apply the relevant subsections of SANS 10142-1			
GENERAL COMMENTS OF OBSERVER:			

Date.....	Time started.....	Time completed.....
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)

LOGBOOK				
Date	Assignment task	Start	Finish	Total Hours
Date	Learner signature	Date	Assessor signature	



LEARNER FILE DIVIDER

CONTENT	CHECKED
PRACTICAL MODULES	

PRACTICALS MODULES

KM ID	TITLE	Credits
PM - 4	Install wireways, NQF Level 4.	5

INSTALL WIRE-WAYS

PRACTICAL MODULE

Module #	671101000-PM-04,
NQF Level	level 4
Notional hours	50
Credit(s)	Cr 5
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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INTRODUCTION

Congratulations on completing the program. As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility**. Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

Acknowledgment of Receipt

I _____ (Learner) acknowledge receipt of my on the job Training Logbook on this the _____ day of _____ 20 ____

The process of on-the-job training has been explained to me.

Signature of Learner

Name of Assessor: _____

Signature of Assessor

Instructions to the Assessor

Congratulations on your appointment as a assessor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
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- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
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- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Responsibilities of the Employer

- Creating an atmosphere conducive to learning
- Giving learners ample access to the working environment. Remember that learners should be productive employees to get practical exposure to all aspects of the transport operation as required by the learnership
- Ensure that learners, mentors and assessors attend all training required and arrange and pay travelling and accommodation costs
- Ensure availability of sufficient mentors and workplace assessors

Training Provider Responsibility

- Provide all practical learning material in electronic and/or hard copy, depending on the circumstances
 - Provide training for mentors and workplace assessors if required and provide the learning material
 - Visit employers to monitor progress and provide guidance and feedback.
 - Provide an online and telephonic support system to all mentors and learners
-

Provider Accreditation Requirements for the Module

Physical Requirements:

Demonstrate access to:

Electrical environment

categories of tools, equipment and processes in order for learners to be able to plan worksite set up for installing, wiring and connecting electrical equipment and control systems

Human Resource Requirements:

Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting, commissioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Trade Certificate.

Trainer/learner ratio 1 to 12

Workshop coach/learner ratio 1 to 4

Legal Requirements:

Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Partner (AQP)

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) requirements

Critical Practical Activities to be Assessed Externally for the Module

None

Exemptions

None

PRACTICAL SKILL MODULE SPECIFICATIONS

List of Practical Skill Module Specifications

671101000-PM-01	Use hand, power and measuring tools,L3,Cr28
671101000-PS-02	Plan worksite set up for installing, wiring and connecting electrical equipment and control systems,L3, Cr5
671101000-PS-03	Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems,L3,Cr3
671101000-PS-04	Install wire-ways,L4, Cr5
671101000-PS-05	Install, wire and connect electrical equipment and control systems, L4,Cr38
671101000-PS-06	Conduct pre-commission inspection (power off) and test new and existing installations, L4,Cr5
671101000-PS-07	Carry out commissioning tests,L4,Cr13
671101000-PS-08	Fault find and repair electrical control systems and electrical installations, L4, Cr22

Purpose of the Practical Skill Module 4

The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The learners will be practicing skills related to the installation of wire-ways

The learner will be required to:

PM-04-PS01: Confirm the selection and installation method of the selected wire ways

PM-04-PS02: Use installation tools and equipment

PM-04-PS03: Design and install wire ways

SECTION A-Guidelines for Practical Skills

PM-04-PS01: Confirm the selection and installation method of the selected wire ways

Scope of Practical Skill

Given work instructions, checklists, work area, activity documents, templates, safety and quality principles and standard operating procedures, the learner should be able to:

- PA0101 Select the correct wireways
- PA0102 Select the method of installation

Applied Knowledge

- AK0101 Relevant subsections of SANS 10142-1
 - AK0102 Installation methods and techniques
 - AK0103 Hazard and risk procedures
-

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Select the correct wire ways according to SANS 10142-1 that meet the operational conditions of the installation.
2. Select the correct installation methods according to SANS 10142-1, the relevant job instruction and environmental requirements.

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Select the correct wire ways according to SANS 10142-1 that meet the operational conditions of the installation.			
2. Select the correct installation methods according to SANS 10142-1, the relevant job instruction and environmental requirements.			
GENERAL COMMENTS OF OBSERVER:			

Date.....	Time started.....	Time completed.....
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)

SECTION B-Guidelines for Practical Skills

PM-04-PS02: Use of installation tools and equipment

Scope of Practical Skill

Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, the learner should be able to:

PA0201 Use Tools, Materials and Equipment

PA0202 Use Personal Protective Equipment

Applied Knowledge

AK0201 Methods and techniques for using personal protective equipment

AK0202 Methods and techniques of using tools

AK0203 Methods and techniques of using power tools

AK0204 Methods and techniques of ensuring workshop / worksite safety tools

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Use Tools, material and equipment and correctly while adhering to worksite and workshop safety procedures.
2. Select and use Personal protective equipment according to statutory requirements.
3. Select and use Power tools correctly and safely while adhering fully to the workshop safety regulations.

OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Use Tools, material and equipment and correctly while adhering to worksite and workshop safety procedures.			
2. Select and use Personal protective equipment according to statutory requirements.			
3. Select and use Power tools correctly and safely while adhering fully to the workshop safety regulations.			
GENERAL COMMENTS OF OBSERVER:			

Date.....	Time started.....	Time completed.....
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)

SECTION C-Guidelines for Practical Skills

PM-04-PS03: Design and install wire ways

Scope of Practical Skill

Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, the learner should be able to:

- PA0301 Design the layout of wireways
- PA0302 Install wire ways
- PA0303 Terminate and join the wireways
- PA0304 Install wire way accessories
- PA0305 Bond wireways

Applied Knowledge

- AK0301 Methods and techniques for using personal protective equipment
 - AK0302 Methods and techniques of using tools
 - AK0303 Methods and techniques of using power tools
 - AK0304 Methods and techniques ensuring workshop / worksite Safety
 - AK0305 Relevant subsections of SANS 10142-1
 - AK0306 Safety and performance standards of wireways
-

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Install Conduit correctly and bent to the correct radius as prescribed in SANS 10142-1
2. Install Cable racks, ladders and trays as prescribed by the manufacturer's specifications
3. Install Trunking and ducting as prescribed by the manufacturer's specifications
4. Use the correct methods for terminating and joining wireways in accordance with the installation requirements.
5. Select the correct methods for the installation of wireways that fully complies with SANS 10142-1.
6. Install the wire ways neatly (vertically, horizontally and offsets parallel and similar) and complies with industry acceptable standard.

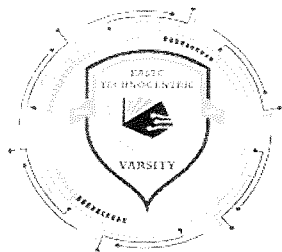
OBSERVATION EVALUATION

NAME: SURNAME: COMPANY: ID:	EVALUATION CHECKLIST DATE: TIME:
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EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Install Conduit correctly and bent to the correct radius as prescribed in SANS 10142-1			
2. Install Cable racks, ladders and trays as prescribed by the manufacturer's specifications			
3. Install Trunking and ducting as prescribed by the manufacturer's specifications			
4. Use the correct methods for terminating and joining wireways in accordance with the installation			

requirements.			
5. Select the correct methods for the installation of wireways that fully complies with SANS 10142-1.			
6. Install the wire ways (vertically, horizontally and offsets parallel and similar) and complies with industry acceptable standard.			
GENERAL COMMENTS OF OBSERVER:			
Date.....	Time started.....	Time completed.....	
ASSESSOR NAME	ASSESSOR SIGNATURE	LEARNER (SIGNATURE)	

LOGBOOK				
Date	Assignment task	Start	Finish	Total Hours
Date	Learner signature	Date	Assessor signature	

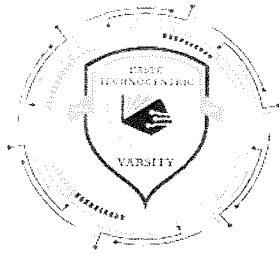


LEARNER FILE DIVIDER

CONTENT	CHECKED
PRACTICAL MODULES	

PRACTICALS MODULES

KM ID	TITLE	Credits
PM - 5	Install, wire and connect electrical equipment and control systems, NQF Level 4.	38

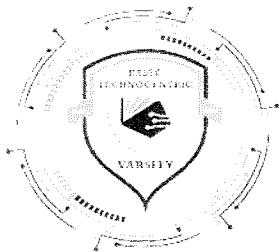


LEARNER FILE DIVIDER

CONTENT	CHECKED
PRACTICAL MODULES	

PRACTICALS MODULES

KM ID	TITLE	Credits
PM - 6	Conduct pre-commission inspection (power on and off) fault find and test new and existing installations, NQF Level 4.	5



LEARNER FILE DIVIDER

CONTENT	CHECKED
PRACTICAL MODULES	

PRACTICALS MODULES

KM ID	TITLE	Credits
PM – 7	Carry out commissioning tests, NQF Level 4.	13

GENERAL

